



UNIVERSITY OF
WATERLOO

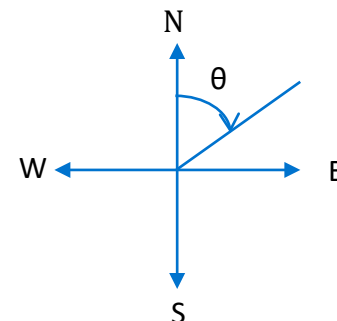
MTE 203 – Week 5 Tutorial

Midterm Exam Fall 2012 – Problem 3

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The elevation z of a hill is described by the function

$$z = f(x, y) = 0.1y^2e^{-x} \quad \text{for } 0 \leq x \leq 2; \quad 0 \leq y \leq 2$$



The compass directions N (north), S (south), E (east), and W (west) are shown in the picture above. The values of x , y , and z are in kilometers. The value of x is the distance eastwards from the origin, and the value of y is the distance northwards from the origin.

- Write down the equation that defines the z contours
- Calculate the maximum slope at point $P(1,1)$ on the hill. From $P(1,1)$, what direction (i.e. what angle in degrees, measured clockwise from N) should a skier follow for the steepest rate of descent?
- From $P(1,1)$ in which direction (i.e. what angle in degrees, measured clockwise from N) is the slope of the surface equal to zero?
- What slope is encountered at $P(1,1)$ if a skier moves from P across the hill directly towards the origin $(0,0)$?

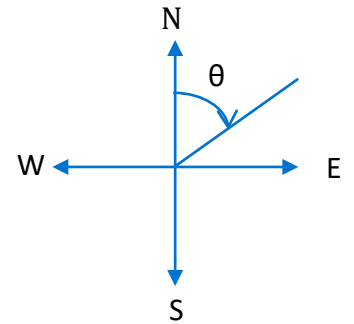


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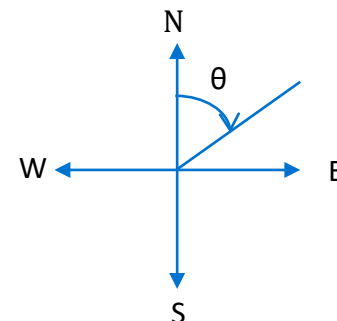
- a. Write down the equation that defines the z contours



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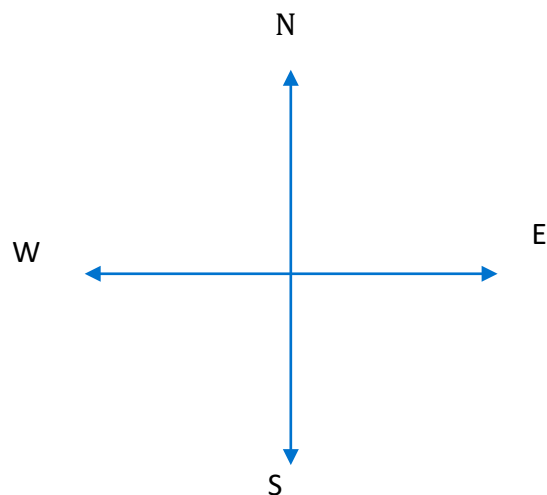
- b. Calculate the maximum slope at point $P(1,1)$ on the hill. From P , what direction (i.e. what angle in degrees, measured clockwise from N) should a skier follow for the steepest rate of descent?



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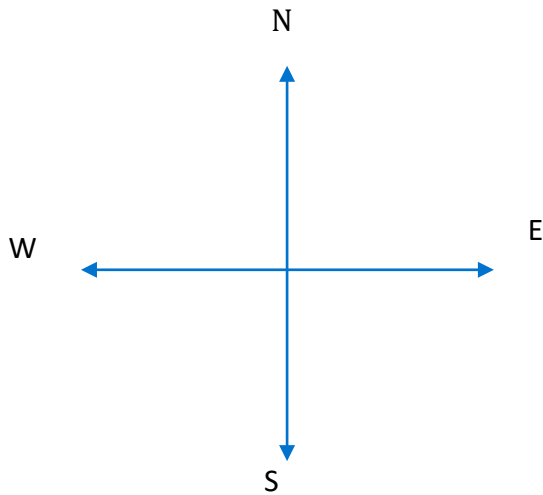
From $P(1,1)$, what direction (i.e. what angle in degrees, measured clockwise from N) should a skier follow for the steepest rate of descent?



- c. From $P(1,1)$ in which direction (i.e. what angle in degrees, measured clockwise from N) is the slope of the surface equal to zero?

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- d. What slope is encountered at $P(1,1)$ if a skier moves from P across the hill directly towards the origin $(0,0)$?



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Thank you

Spring 2026